coupling constant.9 Although the two string theories appear to be unrelated when analyzed using the perturbative approximation scheme, we now see that each transforms into the other—somewhat like the transmutation between water and ice—as their coupling constants are varied in value.

This central new kind of result, in which the strong coupling physics of one theory is described by the weak coupling physics of another theory, is known as strong-weak duality. As with the other dualities discussed previously, it tells us that the two theories involved are not actually distinct. Rather, they give two dissimilar descriptions of the same underlying theory. Unlike the English-Chinese trivial duality, strong-weak coupling duality is powerful. When the coupling constant of one member of a dual pair of theories is small, we can analyze its physical properties using well-developed perturbative tools. If the coupling constant of the theory is large, however, and thus the perturbative methods break down, we now know that we can use the dual description—a description in which the relevant coupling constant is small—and return to the use of perturbative tools. The translation has resulted in our having quantitative methods to analyze a theory we initially thought to be beyond our theoretical abilities.

Actually proving that the strong coupling physics of the Type I string theory is identical to the weak coupling physics of the Heterotic-O theory, and vice versa, is an extremely difficult task that has not yet been achieved. The reason is simple. One member of the pair of the supposedly dual theories is not amenable to perturbative analysis, as its coupling constant is too big. This prevents direct calculations of many of its physical properties. In fact, it is precisely this point that makes the proposed duality so potent, for, if true, it provides a new tool for analyzing a strongly coupled theory: Use perturbative methods on its weakly coupled dual description.

But even if we cannot prove that the two theories are dual, the perfect alignment between those properties we can extract with confidence provides extremely compelling evidence that the conjectured strong-weak coupling relationship between the Type I and Heterotic-O string theories is correct. In fact, increasingly clever calculations that have been performed to test the proposed duality have